

New York University Tandon School of Engineering

Course Outline **CS-GY 6063** Practical Software Engineering I

Fall 2026

Professor: Gennadiy Civil

Section : Software Engineering I

Ananya Agarwal (aa13549@nyu.edu)

CS-GY 6063 I

9/2/2026 - 12/21/2026

6:00 PM - 8:30 PM W

Jacobs Hall, 6 Metrotech Room 775 Loc: Brooklyn Campus

Section: Software Engineering I INET

Brian Cheng (btc4350@nyu.edu)

CS-GY 6063 INET

9/14/2026 - 12/16/2026

7:00 PM - 9:30 PM M

Online Loc: <https://nyu.zoom.us/j/8027951456>

Contact Professor:

Email: gcivil@nyu.edu

Office hours: by appointment

Course Prerequisites:

- 1) Graduate Status
- 2) Solid ability to program in **Python**. Some familiarity with Django is helpful.
Please note: The course **will not** teach how to program in Python. The students are required to be able to do so already. Experience using version control systems (git in particular) is important.

Course Description:

Learn how to conceptualize, build, deploy, maintain and re-factor software in software development teams.

IMPORTANT: Below is an analysis based on the students' responses for the question “**On average, how many hours per week did you spend on course-related work outside of class?**” Use that to set expectations for the class!

Workload Expectations for Software Engineering I

This course is designed to feel like a realistic software project, not just a set of small homework problems. That means you should expect a **steady, meaningful time commitment** and past students have successfully managed it alongside their other courses, jobs, and responsibilities.

Looking at recent semesters of this exact course:

- Students typically spend **around 11–12 hours per week outside of class** on course-related work.
- **About 70%** of students reported spending **9 or more hours per week**.
- **Nearly half** reported spending **12 or more hours per week**.
- Very few students reported only doing a couple of hours: about **1%** reported **0–2 hours per week**.

Most of this time goes into **team project work, coding, debugging, and coordination**, similar to what you would do on a real software team. Students who plan ahead, communicate with their team, and work consistently each week tend to find the load **very manageable and rewarding**.

You should think of this as a **substantial, project-centered course**. If you budget **around 10–15 hours per week** outside of class, you will be in line with how successful students have approached it.

How much time past students actually spent

All sections, all semesters combined, N = 70

Hours/week outside class	Number of students	Share of students
0–2	1	1%
3–5	8	11%
6–8	13	19%
9–11	17	24%
12–14	11	16%

Hours/week outside class	Number of students	Share of students
15–17	11	16%
18–20	5	7%
20+	4	6%

Expectation for you:

Plan for **roughly 10–15 hours/week of focused work outside of class**. Your actual time may be a bit lower or higher depending on your background and your team, but if you spread the work out over the week and stay engaged with your group, this commitment is very manageable and is a key part of what makes the course valuable.

Student Reviews from 2022:

Course: [Software Engineering I](#)

Dec. 14, 2022

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This is a very practical class. The outcome of your project depends on your team, but my team was able to work very well together and the professor is upfront about this component of the class. I would advise brushing up on both django and javascript beforehand because these skills are not taught per se in the class. It's more about gaining experience of working on a software development team.

Course: [Software Engineering I](#)

Oct. 2, 2022

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This isn't a course, its a mini-internship. At the end of the semester you have a working project which you can showcase anywhere and everywhere you want. But the catch is that majority of your marks are dependent on your teammates.

Course: [Software Engineering I](#)

Oct. 2, 2022

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The course is completely practical and almost no theory. I think that i what software engineering should be like: Hands on approach.

Course Objectives:

- Learn how to work as part of the Software Development Team
- Learn full cycle software engineering effort including requirements management, software development management and delivery
- Gain exposure to practical approaches for Agile development
- Gain exposure to continuous integration techniques
- Learn efficient unit testing and test-first development principals

Course Structure

Weekly lectures/labs plus a significant practical hands-on project.

- There will be constant communication expected between the team members within the teams as well as between the teams and the instructor.
- The project work will follow Agile methodology organized into week-long Sprints with clearly identified, written down and reported Sprint goals.
- Each team will be presenting a Sprint demo at the end of each Sprint.
- Students are expected to meet in their respective teams outside the lecture hours for daily or near-daily Sprint daily scrum meetings. The team meetings minutes have to be reported in writing and submitted.

Readings

Required book:

Essential Scrum: A Practical Guide to the Most Popular Agile Process
(Addison-Wesley Signature): A Practical Guide To The Most Popular Agile Process (Addison-Wesley Signature Series (Cohn))

Optional book, highly recommended:

Software Engineering at Google: Lessons Learned from Programming Over Time - by Hyrum Wright, Titus Winters, and Tom Manshreck

Course requirements

Very few widely-used products are built by an individual, so students must learn to collaborate with others both in design and construction of the codebase. Software engineering is a team sport.

Practical Software Engineering classes **require** heavy teamwork participation. A student wishing to work “on their own” will **not** be able to satisfactorily complete the class.

The class will be graded as follows:

- 45% - successful completion of the team project.
 - Format: Working software accessible over the Internet.
 - All project code deployed to GitHub
 - All project code complies with established Code Style guidelines
 - Special emphasis on the test suite completeness and test code coverage percentage
- 45% - Teamwork
 - 360 Team mutual evaluation forms evaluating participation: (90%)
- 10% Individual Homework Assignments

NOTE: Individual Homework Late Submission Policy:

OnTime, working HW=100 points

Every hour of being late = -1 point.

Every late day = -25 points. (I know there are 24 hours in a day).

Part I: Introduction to Agile Development in Teams.

Note: All Listed Assignments are due prior to the next class unless otherwise noted

Class 1: Introduction and Team Formation

Pre-Requisites review, Required software overview, Process overview, Teams Formation, Ideas for team projects. Grading overview, emphasis on teamwork.

- Assignment: Environment Checklist - required accounts (GitHub, Slack, etc) and all software environment
- Assignment: Team communications established, chat rooms, shared documents repository.

- Assignment: Each Team started and shared with the instructor “team project ideas” document. The document needs to list 2-3 ideas for your team projects.

Class 2: Just enough Agile Process to get started.

Agile development overview. Scrum process overview and essential Scrum ceremonies. Directions for student teams. Daily Scrum meeting reporting requirements. What is the role of Scrum Master? What is the Definition of Done.

- Lab: Daily scrum meeting play-acting
- Class Review: Project Ideas, Class Discussion
- Reading: Chapter 1 Introduction
- Reading: Chapter 2 Scrum Framework
- Reading: Chapter 3 Agile Principles
- Assignment: Sprint 1:
 - Sprint Goal: Finish and submit proposals for the team projects. The proposals are expected to be complete documents rather than brief ideas from the week before. Teams are expected to vote internally and submit two-three proposals per team. The team project document is expected to show team collaboration and discussion trail
 - Assignment: Team daily scrum meetings conducted by the team and reports submitted. *NOTE: From now on each Sprint has to include daily scrum meetings reports as a deliverable every week.*

Class 3: First Sprint Demo. How to write user Stories. What is the role of Product Owner.

- Instructor will approve 1 project per team prior to the class
- (Lab) Sprint Demos, all teams. Teams present their projects.
- (Lecture) How to Write User Stories
- Reading: Chapter 4: Sprints
- Reading: Chapter 5, Requirements and User Stories
- Reading: Chapter 6: Product Backlog
- Assignment: Sprint 2:
 - Sprint Goal: Produce requirements for team projects. The team is expected to collaborate, review documents and produce agreed-upon requirements in form of user stories.

Part II: Practical Software Delivery

Class 4: Let's design our projects.

- (Lab) Sprint Demos, all teams. Review user stories. Class discussion

- (Lecture) Initial System design for team projects. Potentially useful UML diagrams explained
- Assignment: Sprint 3:
 - Sprint Goal:
 - Produce design document for the team's projects. Use UML as appropriate, explain why the team chose to use or not to use UML
 - As always team participation and design discussions are expected
- Individual assignment-Django Hello Word developed and deployed. All students are expected to complete and submit individual Code Repositories on GitHub
- Reading: UML sequence diagram explained:
<https://www.geeksforgeeks.org/unified-modeling-language-uml-sequence-diagrams/>
- Reading: Django Tutorial.
<https://docs.djangoproject.com/en/4.1/intro/tutorial01/>

Class 5: Let's write some code already!

- (Lab) Sprint Demos, all teams. Review project design documents. Class discussion
- (Lecture) Sprint Planning. Agile Poker.
- (Lab) Sprint Planning. Agile poker.
- (Lecture) Django framework overview and how it fits into the project design.
- Assignment: Sprint 4:
 - Sprint Goal:
 - Chosen User Stories documented
 - Pending user stories documented in the project backlog
 - Write code to satisfy user stories chosen for Sprint
 - Deploy Sprint 04 solution to be available for Sprint Demo review
 - First Sprint Retrospective team discussion and report.
NOTE: From now on, in addition to daily scrum meetings report, the team retrospective report is due prior to the next week class.
- Reading: Django Documentation. <https://docs.djangoproject.com/en/4.1/>

Class 6: Continuous Integration and the Role of Test Suites in team development

- (Lab) Sprint Demos, all teams. Teams present what they coded. Class discussion on team retrospectives and team backlogs
- (Lecture) Introduce Continuous Integration.
- (Lecture) Team Velocity and its proper use

- (Lecture) Coding Standards, integrating Python linter and formatter into development process
 - Assignment: Sprint 5:
 - Sprint Goal:
 - a) All team members set up continuous integration for their respective individual Hello World Django projects.
 - b) All team members set up Initial Test Suites for their respective individual Hello World Django projects.
- NOTE: These are individual goals however the teams are still expected to meet, discuss and help each other out. Each team is responsible for all team members. No one is left behind.
- c) Team project development continues
 - Reading: Travis CI Tutorial, <https://docs.travis-ci.com/user/tutorial/>

Class 7: Why what has been developed so far is not good enough and how we will make it better using Continuous Integration and Test Suites

- (Lab) Sprint Demos, all teams.
- (Lecture) Introduce Test-Driven Development.
- (Lecture) Introduce Code Coverage as a metric for Test-Driven Development.
- Assignment: Sprint 6:
 - Sprint Goal:
 - Refactor code for previous Sprint - Write unit tests, make unit tests pass, Check code coverage. Change the project skeleton if needed.
 - Add Continuous Integration to the team's project GitHub repositories
 - Ensure that the Test Suites are located in the recommended directories
- Reading: Testing in Django.
<https://docs.djangoproject.com/en/4.1/topics/testing/>

Class 8: Project Work continues.

- (Lab) Sprint Demos, all teams. Teams present selected test suites - 1 per team. Class discussion.
- Assignment: Sprint 7:
 - Sprint Goal: Test suite passing for Sprint goal. Stories TBD as per Sprint Planning

Class 9: Project Work continues.

- (Lab) Sprint Demos, all teams -- **MIDTERM PRESENTATION**
- Assignment: Sprint 8:
 - Sprint Goal: Test suite passing for Sprint goal. Stories TBD as per Sprint Planning

Class 10: Project Work continues.

- (Lab) Sprint Demos, all teams
- Assignment: Sprint 9:
 - Sprint Goal: Test suite passing for Sprint goal. Stories TBD as per Sprint Planning

Class 11: Project Work continues.

- (Lab) Sprint Demos, all teams
- Assignment: Sprint 10:
 - Sprint Goal: Test suite passing for Sprint goal. Stories TBD as per Sprint Planning

Class 12: Project Work continues.

- (Lab) Sprint Demos, all teams
- Assignment: Sprint 11:
 - Sprint Goal: Test suite passing for Sprint goal. Stories TBD as per Sprint Planning

Class 13: Project Work continues.

- (Lab) Sprint Demos, all teams
- Assignment: Sprint 12:
 - Sprint Goal: Test suite passing for Sprint goal. Stories TBD as per Sprint Planning

Class 14: Project Work continues.

- (Lab) Sprint Demos, all teams
- Assignment: Sprint 13:
 - Sprint Goal: Test suite passing for Sprint goal. Stories TBD as per Sprint Planning

Class 15, last class, all projects due

- (Lab) Project Demos, all teams
- Make a GitHub release version 1.0
- Class discussion

Class16, No Class: All grades are final.

Moses Center Statement of Disability

If you are a student with a disability who is requesting accommodations, please contact New York University's Moses Center for Students with Disabilities (CSD) at [212-998-4980](tel:212-998-4980) or mosescsd@nyu.edu. You must be registered with CSD to receive accommodations. Information about the Moses Center can be found at www.nyu.edu/csd. The Moses Center is located at 726 Broadway on the 3rd floor.

NYU School of Engineering Policies and Procedures on Academic Misconduct – complete Student Code of Conduct [here](#)

- A. Introduction: The School of Engineering encourages academic excellence in an environment that promotes honesty, integrity, and fairness, and students at the School of Engineering are expected to exhibit those qualities in their academic work. It is through the process of submitting their own work and receiving honest feedback on that work that students may progress academically. Any act of academic dishonesty is seen as an attack upon the School and will not be tolerated. Furthermore, those who breach the School's rules on academic integrity will be sanctioned under this Policy. Students are responsible for familiarizing themselves with the School's Policy on Academic Misconduct.
- B. Definition: Academic dishonesty may include misrepresentation, deception, dishonesty, or any act of falsification committed by a student to influence a grade or other academic evaluation. Academic dishonesty also includes intentionally damaging the academic work of others or assisting other students in acts of dishonesty. Common examples of academically dishonest behavior include, but are not limited to, the following:
 - 1. Cheating: intentionally using or attempting to use unauthorized notes, books, electronic media, or electronic communications in an exam; talking with fellow students or looking at another person's work during an exam; submitting work prepared in advance for an in-class examination; having someone take an exam for you or taking an exam for someone else; violating other rules governing the administration of examinations.
 - 2. Fabrication: including but not limited to, falsifying experimental data and/or citations.
 - 3. Plagiarism: intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise; failure to attribute direct quotations, paraphrases, or borrowed facts or information.
 - 4. Unauthorized collaboration: working together on work meant to be done individually.

5. Duplicating work: presenting for grading the same work for more than one project or in more than one class, unless express and prior permission has been received from the course instructor(s) or research adviser involved.
6. Forgery: altering any academic document, including, but not limited to, academic records, admissions materials, or medical excuses.

NYU School of Engineering Policies and Procedures on Excused Absences – complete policy [here](#)

- A. Introduction: An absence can be excused if you have missed no more than 10 days of school. If an illness or special circumstance has caused you to miss more than two weeks of school, please refer to the section labeled Medical Leave of Absence.
- B. Students may request special accommodations for an absence to be excused in the following cases:
 1. Medical reasons
 2. Death in immediate family
 3. Personal qualified emergencies (documentation must be provided)
 4. Religious Expression or Practice

Deanna Rayment, deanna.rayment@nyu.edu, is the Coordinator of Student Advocacy, Compliance and Student Affairs and handles excused absences. She is located in 5 MTC, LC240C and can assist you should it become necessary.

NYU School of Engineering Academic Calendar – complete list [here](#).

Also, please pay attention to notable dates such as Add/Drop, Withdrawal, etc. For confirmation of dates or further information, please contact Susana: sgarcia@nyu.edu